# Scaling the Attack Breaks Defense Containment in Task-Level Radar–Jammer Self-Play

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***Abstract*—Multifunction phased-array radar networks must schedule heterogeneous, deadline-constrained missions while facing coordinated, adaptive interference, and link-level signal-to-interference ratios do not by themselves reveal whether the network completes its mission load. We present a task-level adversarial evaluation protocol, FluxPhased, that couples deadline-constrained mission scheduling to bearing-aware planar-array link physics, and use it to measure how the radar–jammer balance shifts when the attack is scaled up. Missions carry service and bearing identities, deadlines, and exact detection credit; the link model combines generalized array-factor responses with finite jammer energy budgets. A pre-training contestability test and a four-view evaluation protocol separate natural scheduling loss from adversarial loss, and evaluation-only scripted baselines anchor both learned teams against non-learning behavior.**

**The system-level finding is stated in engineering terms: with the total jammer activation budget held fixed at 63 cells per episode, distributing the attack across two spatially separated jammers reduces the fraction of adversarial mission denial that learned radar scheduling removes by roughly two-thirds. In the one-jammer versus two-radar game, the valid 12-dB S6 runs neutralize $63.7\%\pm1.0\%$ of the jammer's floor-adjusted disruptive power across two training seeds. When the attacker team is doubled under the same budget, three converged seeds yield a stable 2v2 equilibrium with head-to-head mission drop $0.3366\pm0.0143$ and neutralization $23.0\% \pm 1.1\%$. Radar competence against idle jammers remains low, showing that the degradation is an offensive scaling effect rather than a corresponding defender collapse. A co-located-jammer control attributes most of the change to attacker count under the S7 separated-radar geometry (neutralization $28.4\%$), with cross-fire geometry contributing a further reduction to $24.2\%$. Evaluation-only scripted baselines show that both learned teams dominate random and fixed-stare policies, yet a greedy mission-stare radar holds drop at $0.0889$ against every trained jammer team, so head-to-head values are equilibrium properties of the co-adapted pair rather than performance ceilings. Finally, a lightweight opponent-class mixture reduces singleton relative leverage from $0.502$ to $0.262$ at 75% singleton exposure, with the largest neutralization value at 50% exposure. The protocol establishes a measurable separation between equilibrium stability, attack scaling, and opponent-distribution robustness for learned radar resource management under electronic attack.**

***Index Terms*—Radar resource management, multifunction radar, phased arrays, electronic warfare, electronic attack, mission scheduling, multi-agent reinforcement learning, self-play, adversarial robustness.**

## I. Introduction

Electronic-support and electronic-attack systems interact through a resource-constrained loop: an interferer chooses where and when to spend energy, while a multifunction radar chooses which service and bearing to observe before each mission deadline. A high instantaneous JNR can be operationally irrelevant if the radar detects the corresponding target in another dwell, whereas a modest interference pulse can matter when it arrives at the only vulnerable point in a deadline window. The operational outcome is therefore mission completion, not a single link-budget scalar.

This distinction exposes a gap between two established lines of work. Radar–jammer learning studies formulate waveform, frequency, or power decisions as sequential games. Multifunction-radar scheduling studies optimize scan and track allocation under time or power constraints. These lines of work leave open a system-level question: how does the balance of an adaptive game change when the number of coordinated attackers increases while the defender's mission workload and total interference budget remain fixed?

We answer this question with a controlled S6-to-S7 progression in FluxPhased. S6 contains one learned jammer and two learned radar heads. S7 contains two parameter-shared jammer agents and two parameter-shared radar agents. The S7 team receives a 63-cell activation budget, split 32/31 between jammers. This controls total jammer activation while the radar-site geometry changes from S6 to S7; the matched co-located-jammer control isolates the additional cross-fire component within the S7 geometry.

The main result is a change in containment, not a corresponding failure of learning. Across three S7 seeds trained to 3000 iterations, head-to-head mission drop is $0.3366\pm0.0143$, compared with $0.0888 \pm 0.0053$ in the valid two-seed 12-dB S6 baseline. The floor-adjusted fraction of marginal jammer disruption removed by learned radars falls from $63.7\%\pm1.0\%$ to $23.0\%\pm1.1\%$. The radar floor against idle jammers remains low ($0.0194\pm0.0008$ in S7 versus $0.0275\pm0.0110$ in valid S6), and the S7 trajectory is stationary from approximately iteration 1700 through 3000. Thus, the second attacker changes the stable equilibrium's scale while preserving low idle-jammer loss and training stability.

Evaluation-only scripted baselines bound these equilibrium numbers. Both learned teams clearly beat random and fixed-stare scripted opponents, but a greedy mission-stare radar holds mission drop at $0.0889$ against every trained jammer team—below the learned head-to-head value and below the natural scheduling floor. The head-to-head drop is therefore an equilibrium property of the co-adapted pair, not a ceiling on radar performance, and both teams are specialized to their training opponents.

This paper makes five contributions:

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1) We formulate a task-level adversarial evaluation protocol for multifunction radar networks under electronic attack, in which deadline-constrained mission loss is coupled to array-factor link physics, finite energy, and two-team multi-agent PPO. The protocol, not the specific learner, is the reusable artifact.
2) We quantify a benchmark-level attacker-count scaling effect: with total jammer activation controlled, doubling attackers reduces the mission-denial containment of learned radar scheduling by roughly two-thirds, with replication across three converged S7 seeds and a two-seed valid S6 reference.
3) We use a matched co-located-jammer control to decompose the mechanism under the separated S7 radar geometry. Two jammers already reduce containment to $28.4\%$; cross-fire adds a secondary reduction to $24.2\%$.
4) We anchor both learned teams with evaluation-only scripted baselines. Both teams dominate random and stare policies, but a greedy mission-stare radar is unexploited by every trained jammer team, showing that head-to-head drop is an equilibrium quantity rather than a radar performance ceiling.
5) We test a minimal opponent-class mixture. The singleton-to-pair disruption ratio falls from $0.502$ to $0.262$ as singleton exposure increases, identifying opponent-class mixing as a practical robustness intervention while documenting its training-budget confound.

The rest of the paper defines the benchmark and evaluation metric, presents convergence and scaling results, tests the count–geometry decomposition, analyzes opponent-specific adaptation, and evaluates the lightweight mixing intervention. We use "drop" for the fraction of eligible missions that are not completed before their deadlines; higher drop therefore indicates greater jammer-side success.

## II. Related Work

### *A. Cognitive radar, sensor management, and resource scheduling*

Cognitive radar frames the sensing loop as an adaptive decision process with feedback from the receiver to the transmitter [1]. The fully adaptive radar (FAR) framework formalized perception–action cycles for detection and tracking [2], [3], and later work developed cognitive resource management from adaptive scheduling toward learned allocation [4]–[6], including game- and bandit-based approaches to radar coexistence and adversarial settings [7], [8]; the broader sensor-management literature treats the same attention-allocation problem across sensing systems [9]. These lines establish that a radar's transmit behavior should adapt to its environment, but the environment is usually non-adaptive or drawn from a fixed distribution; the opponent does not learn against the radar. Reinforcement learning has likewise been applied to phased-array beam and dwell scheduling under overload [10], most recently with model-based deep RL for cognitive multifunction radar task scheduling in this Transactions [11]—again with scripted demand and no learning adversary. Our protocol keeps this resource-management structure—services, bearings, deadlines, and a finite activation budget—but places a learning jammer team on the other side of the link.

### *B. Learning in radar–jammer and anti-jamming games*

A growing literature learns jamming and anti-jamming decisions as games. Multi-agent deep RL has been used for cognitive-radar anti-jamming decision-making [12], [13] and for multi-jammer radar countermeasures [14]; the wireless anti-jamming literature is surveyed in [15], with deep-RL power and channel defenses dating back to two-dimensional jamming avoidance [16]. Neural fictitious self-play provides a standard solution concept for such imperfect-information games [17]. The dominant endpoints in this literature are link-level quantities: SINR, detection probability at the current dwell, or a per-step utility. FluxPhased complements this line with a mission-level endpoint: a mission succeeds only when the correct service detects the correct bearing within a deadline window, so per-step link wins do not immediately translate into task denial. This distinction is what makes containment—the fraction of adversarial marginal disruption that a defense removes—measurable at all.

### *C. Multi-agent reinforcement learning and self-play*

Our trainer combines standard components: PPO with clipped surrogate objectives [18], parameter-sharing multi-agent actors with centralized critics in the CTDE family [19]–[23], and self-play [17], [24], [25]. The literature is explicit that learning with a learning opponent creates non-stationarity and opponent-specialization problems [26], [27] and that differentiable games can have cyclic or non-transitive dynamics whose equilibria require game-theoretic rather than purely gradient tools [28], [29]. Population and league methods address this by diversifying opponents [24], [29]; role-oriented and macro-strategy methods structure within-team coordination [30], [31]. We deliberately use none of these stronger machineries: the S7 game converges under plain two-team self-play, which isolates attacker-count scaling from league effects, and R5 tests only a minimal opponent-class mixture. The singleton-vulnerability result of Section V shows why opponent-population questions remain relevant for this benchmark even after convergence.

### *D. Position of this work*

FluxPhased sits between the radar resource-management community and the adversarial-learning community. Its action space is link-facing (cells, beams, services), but its reported endpoint is task completion under service, bearing, deadline, and energy constraints. This enables a question that neither link-level anti-jamming studies nor single-sided scheduling studies answer: whether an apparent defensive advantage survives attacker-count scaling, and whether the resulting two-team equilibrium is stable or cyclic. The comparison is built around three controls—the idle-radar floor, a scripted-radar jammer view, and a co-located-jammer ablation—plus evaluation-only scripted baselines (random, greedy, sweep, and stare policies) that anchor the learned policies against non-learning behavior. The controls separate absolute loss, adversarial marginal loss, equilibrium stability, robustness to a changed opponent class, and the distinction between an equilibrium policy and a mission-greedy policy.

## III. Benchmark and Method

### A. Task-level mission model

At each time step, each service can receive an arrival. An admitted mission is represented by

$$m = (s, a, t_{\mathrm{arr}}, t_{\mathrm{dl}}, n_{\mathrm{det}}), \tag{1}$$

where $s$ is the service, $a \in \{0, \ldots, 4\}$ is its azimuth sector, and the deadline is at most six steps after arrival. A mission succeeds when the radar team obtains one detection with the exact $(s, a)$ identity before the deadline. The tracker credits the oldest incomplete mission with the matching service and bearing; detections at the deadline are processed before finalization. The mission drop ratio counts timeouts, admission rejects, and horizon failures over eligible missions:

$$d = \frac{N_{\mathrm{to}} + N_{\mathrm{rej}} + N_{\mathrm{hz}}}{N_{\mathrm{elig}}}. \tag{2}$$

A larger value means that the jammer team denied more missions. We retain completed-but-not-yet-finalized missions in the queue, which makes the deadline accounting explicit and auditable.

### B. Bearing-aware link physics

Each agent selects a beam from a $5 \times 5$ azimuth–elevation grid. A jammer selects a Bernoulli mask over 25 array cells and a categorical beam. For a steering direction $(u_b, v_b)$ and target $(u_t, v_t)$, the normalized planar-array response is separable and follows the standard uniform-planar-array power pattern for a $5 \times 5$ element grid:

$$\begin{aligned} G_{\mathrm{AF}}(b, t) = 10 \log_{10}\left(\frac{A_5(u_t - u_b)^2}{5^2}\right) \\ + 10 \log_{10}\left(\frac{A_5(v_t - v_b)^2}{5^2}\right), \end{aligned} \tag{3}$$

where $A_5$ is the finite-array element factor response evaluated with the same float32 implementation for all experiments; the separable form is the usual product of azimuth and elevation linear-array factors [32], [33]. For jammer $k$ and radar $r$, the received JNR includes jammer cell aggregation, transmit and receive array gains at their pair bearing, service-dependent path loss and receiver gain, and polarization loss. Independent jammers are combined as incoherent power:

$$\mathrm{JNR}_r = 10 \log_{10} \sum_k 10^{\mathrm{JNR}_{kr}/10}. \tag{4}$$

The effective SNR is

$$\mathrm{SNR}_{\mathrm{eff},r} = 10 \log_{10}\left(\frac{10^{\mathrm{SNR}_0/10}}{1 + 10^{\mathrm{JNR}_r/10}}\right) + G_{\mathrm{coh}}, \tag{5}$$

with the idle case defined by $\mathrm{JNR}_r = -\infty$. A mission at azimuth $a$ is detected with

$$p_r(s, a) = \sigma\left(\frac{\mathrm{SNR}_{\mathrm{eff},r} + G_{\mathrm{target}}(b_r, a) - \tau}{w}\right), \tag{6}$$

TABLE I
Link-budget and detection parameters.

| Parameter | Value |
|---|---|
| Service 0 | $f_c = 10.0$ GHz, $B = 10$ MHz, $G_{\mathrm{rx}} = 35$ dB |
| Service 1 | $f_c = 10.5$ GHz, $B = 20$ MHz, $G_{\mathrm{rx}} = 30$ dB |
| Jammer power per cell | $P_{\mathrm{jam}} = 0.1$ W |
| Jammer Tx gain | 20 dB |
| Jammer–radar range | $d = 8$ km (floor 100 m) |
| Polarization loss | 3 dB |
| Noise figure | 5 dB |
| Coherent gain | $G_{\mathrm{coh}} = 20$ dB |
| Baseline SNR | $\mathrm{SNR}_0 = 12$ dB |
| Threshold / width | $\tau = 15$ dB, $w = 3$ dB |

where $\tau = 15$ dB and $w = 3$ dB. This couples radar pointing to both useful detection gain and exposure to interference.

Table I lists the complete link-budget parameterization. The per-pair received JNR follows the standard one-way interference budget $\mathrm{JNR}_{kr} = P_{\mathrm{tx}} + G_{\mathrm{tx}} + G_{\mathrm{rx}} - L_{\mathrm{path}} - L_{\mathrm{pol}} + 10 \log_{10}(\text{spectral overlap}) - N$ with free-space path loss $L_{\mathrm{path}} = 20 \log_{10}(4\pi d/\lambda)$ and noise $N = -174 + 10 \log_{10} B +$ NF [34]; it omits atmospheric and rain terms, which are small at X-band over 8 km relative to the dB-scale margins studied here.

Two combining assumptions deserve justification. First, *within* one jammer, active cells are aggregated with phase-aligned contributions, i.e., the cell mask acts as idealized transmit beamforming; this isolates the scheduling question (which cells, which beam) from array calibration error. Second, *across* jammers, received powers are summed incoherently in (4). The two jammer sites are separated by a baseline of approximately $2d \sin 60^\circ \approx 13.9$ km, i.e., about $5 \times 10^5$ wavelengths at 10 GHz; without phase coordination at that scale, the received phase difference is uniformly uncertain, so expected received powers—not fields—add. Coherent cross-jammer combining would require phase synchronization far beyond the threat model assumed here and would only strengthen the attacker; the incoherent model is therefore the conservative choice for the defense-side conclusions. The jamming waveform is broadband noise with rectangular spectra, so service-dependent rejection enters only through the spectral-overlap factor; we do not model pulse-to-pulse agility, Doppler, or MTI filtering, and Section VII discusses this boundary.

### C. S6 and S7 games

S6 uses one jammer against two co-located learned radar heads. S7 uses two parameter-shared jammers at azimuths $+60^\circ$ and $-60^\circ$ and two radar heads at $+20^\circ$ and $-20^\circ$. Both stages use the validated $\mathrm{SNR}_0 = 12$ dB regime and $P_{\mathrm{jam}} = 0.1$ W per active cell. The S7 team receives a total 63-token budget, split 32/31 between jammers; the S6-to-S7 comparison therefore changes attacker count and radar-site geometry while holding total S7 jammer activation fixed.

Each side uses a parameter-shared multi-head actor. Radar agents choose beam and service; jammer agents choose cell mask and beam. Table II summarizes the action and observation spaces. A centralized privileged critic receives the concatenated public observations of its team during training,

TABLE II
ACTION AND OBSERVATION SPACES.

| | Jammer agent | Radar head |
|---|---|---|
| Actions | cell mask (25 Bern.)<br>beam (25-way) | beam (25-way)<br>service (2-way) |
| Energy | 32/31-token budget | unconstrained |
| Obs. dim | 67 | 60 |
| Obs. | own energy,<br>radar beams/svcs/<br>detected flags,<br>pending map,<br>partner channel | pending $(s, a)$ map,<br>own+partner beams<br>and services,<br>per-jammer DOA<br>and active flags |
| Privileged | team cat. (134) | team cat. (120) |

TABLE III
TRAINING HYPERPARAMETERS (BOTH TEAMS, BOTH STAGES).

| Parameter | Value |
|---|---|
| Environments $\times$ horizon | $16 \times 64$ |
| Actor / critic LR | $3 \times 10^{-5}$ / $10^{-3}$ |
| Discount $\gamma$ / GAE $\lambda$ | 0.99 / 0.95 |
| PPO clip / grad clip | 0.2 / 0.5 |
| Epochs / minibatch | 4 / 256 |
| Target KL (rollback) | 0.02 |
| Entropy coef. (cell, beam, svc) | $2{\times}10^{-2}$, $5{\times}10^{-3}$, $10^{-2}$ |
| Entropy anneal frac. (c,b,s) | 0.7, 0.9, 0.5 |
| Priv. value / distill coef. | 0.5 / 0.1 |
| Value loss coef. | 0.5 |
| Iterations (stage 1 / 2) | 1000 / 2000 (frozen) |

while the deployable actor receives only its local observation and last-action ESM channels. PPO uses generalized advantage estimation, clipped likelihood ratios, per-head entropy schedules, and separate actor, local critic, and privileged-critic optimizers [18], [19]. Rollouts flatten agent slots in agent-major order so the shared team reward is duplicated without mixing trajectories.

### D. Evaluation and containment

We evaluate four views for S7: head-to-head (h2h), learned jammers against scripted sweep radars (jvs), learned radars against idle jammers (rad-idle), and one learned jammer against the pair-trained radars (j1-only). The S6 baseline defines h2h, jvs, rad-idle, and the natural floor; it does not contain the S7-specific j1-only control. The scripted sweep against idle jammers supplies the natural floor $d_{\text{floor}}$. Let $d_{\text{h2h}}$, $d_{\text{jvs}}$, and $d_{\text{idle}}$ denote the corresponding drop ratios.

$$\eta = 1 - \frac{d_{\text{h2h}} - d_{\text{idle}}}{d_{\text{jvs}} - d_{\text{floor}}}. \tag{7}$$

This metric measures the fraction of the jammer's marginal disruption that learning radars remove; it avoids crediting the radar for losses caused by its own scheduling floor. Three properties matter for interpretation. $\eta = 1$ when the learned radars push the head-to-head drop back to their own idle-jammer drop; $\eta = 0$ when the marginal adversarial loss equals the full scripted-radar exposure, i.e., the learned radar removes none of it; values between 0 and 1 are directly readable as removed fractions. The denominator $d_{\text{jvs}} - d_{\text{floor}}$ is the game's realizable disruption range, so $\eta$ is comparable across games only when that range is similar—we therefore always report it alongside the raw drops. Finally, $\eta$ is an equilibrium quantity: it measures containment by the co-adapted radar policy, not the best achievable containment against a frozen opponent (Section V-C). Final evaluations use 64 fixed validation scenarios and three independent action seeds (4242, 777, and 31337).

### E. Pre-training contestability gate

Before training, we enumerate beam responses for each mission azimuth. The gate requires at least three of five azimuths to have best-response detection probability in $(0.05, 0.95)$, a minimum above 0.02, and a maximum above 0.8. The S7 cross-fire profile is $(0.837, 0.912, 0.989, 0.993, 0.855)$; all five sectors remain reachable and three remain contestable. The co-located-jammer control profile is $(0.921, 0.823, 0.989, 0.996, 0.991)$, which predicts a more defense-favorable game before any policy update.

## IV. Experimental Protocol

### A. Training and checkpoints

We use 64 fixed training scenarios and a separate 64-scenario validation manifest. Each environment rollout has 16 parallel environments and a 64-step horizon. The S7 main experiments use three random seeds (20260801, 20260802, and 20260803). Each seed trains for 1000 iterations with the normal per-head entropy schedule and continues to 3000 iterations with entropy coefficients frozen at their minimum. The continuation isolates additional optimization from a second exploration schedule. Table III lists the shared training configuration.

The primary S6 comparison uses the two valid 12-dB training seeds 20260730 and 20260731. An earlier S6 seed, 20260729, was trained accidentally with a 22-dB override; we exclude it from the primary comparison and retain it only as a documented regime check. The valid two-seed S6 aggregate is used wherever an S6 uncertainty is reported.

### B. Mechanism ablation

The co-located-jammer control uses seed 20260811, the same 1000+1000 two-stage protocol, the same 63-token team budget, radar sites at $+20°$ and $-20°$, and both jammer sites at $+60°$. Its corrected evaluation passes the explicit `+60,+60` geometry to the evaluator. The first evaluation generated with the default cross-fire geometry is retained as an audit file and is not used in any reported result.

### C. Opponent-class mixture

R5-lite trains one seed for each singleton exposure fraction $q \in \{0.25, 0.50, 0.75\}$, for 2000 iterations under the same two-stage protocol. On a fraction $q$ of iterations, jammer 1 is forced idle and the radar team updates against the singleton opponent; the jammer-team update is skipped on those iterations. This is a minimal opponent-class mixture rather than a full population league. The reference $q = 0$ is the S7 seed-01 2000-iteration

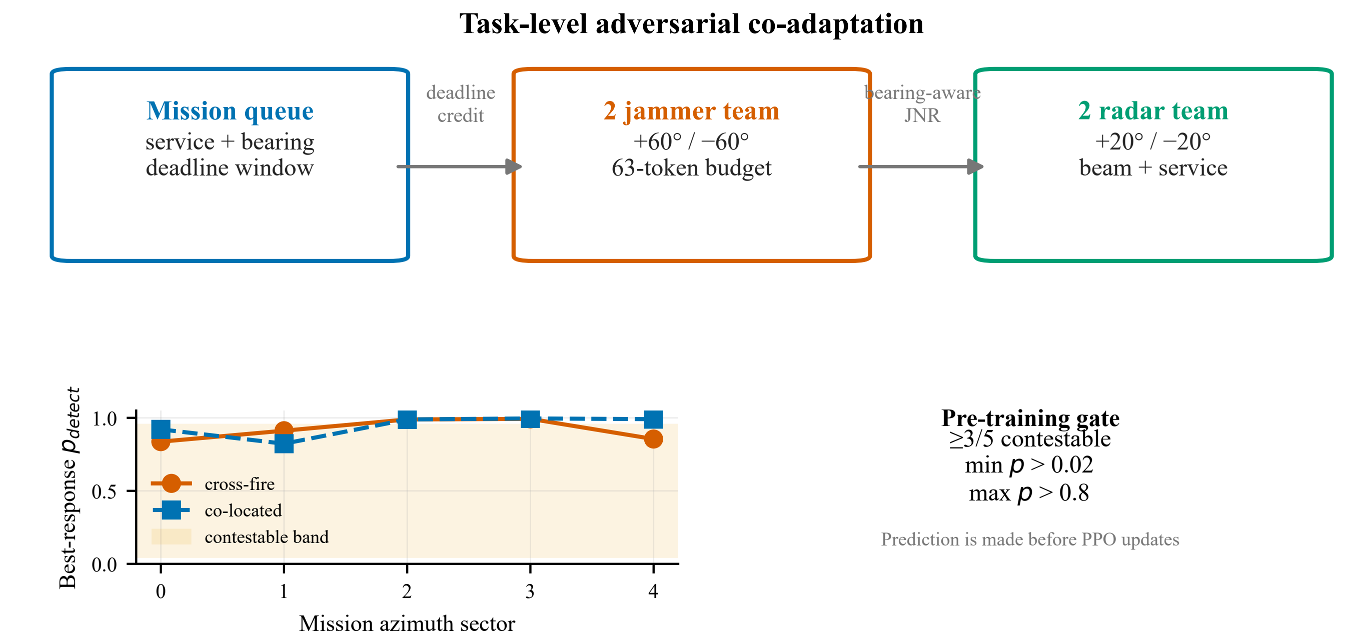


Fig. 1. Benchmark anatomy and pre-training contestability. FluxPhased couples a deadline-constrained mission queue to bearing-aware jammer–radar link physics. The lower panel reports the best-response detection profiles used to gate the cross-fire and co-located configurations before training.

checkpoint. Because jammer updates are skipped on singleton iterations, the absolute jammer strength changes with $q$. We therefore treat $j1/jvs$, $h2h/jvs$, and $\eta$ as the primary cross-condition readouts and present absolute drops as descriptive context.

### *D. Statistical reporting*

For S7, the final evaluation runs four views over all 64 validation scenarios with one repetition for each of three action seeds. The S6 implementation supplies h2h, jvs, rad-idle, and the natural floor but does not supply the S7-specific j1-only control. We report the mean and standard deviation across action seeds for each training seed. Aggregate S7 values report the mean and standard deviation across the three training seeds. The natural floor is computed using scripted radar sweeps against idle jammers under the same environment regime. We retain all raw JSONL logs and final evaluation JSON files in the repository artifact tree.

### *E. Evaluation-only scripted baselines*

Each converged checkpoint is additionally evaluated against five scripted baselines that require no training. The random radar samples beam and service uniformly from the legal sets; the greedy mission-stare radar points each head at the pending service–azimuth cell with the most missions, read from the radar observation; the earliest-deadline-first (EDF) radar points each head at the pending mission with the earliest deadline—unlike the others this is an oracle heuristic, because deadline state is not part of the radar observation; the random jammer activates one uniformly random legal cell and beam per step while energy remains; the stare jammer holds the horizon-plane beam closest to its paired radar-site bearing with one active cell until the budget is spent. Stare beam indices are computed from the site geometry rather than tuned. These views use the same 64 validation scenarios and three action seeds, so they are directly comparable with the learned views. They anchor both teams against non-learning behavior and expose whether the self-play policies are specialized to their training opponents.

### *F. Implementation checks*

The S7 environment and physics tests contain 18 contract and physics gates. They cover action shapes, energy accounting, exact service–bearing credit, generalized array-factor consistency, two-source JNR summation, detection monotonicity, observation channels, ledger identity, and pre-training contestability. Atomic checkpoints and a max-iteration resume criterion protect long runs from interrupted sessions. These engineering checks are separate from the scientific result and are reported to make the training trajectory reproducible.

## V. Results

### *A. Self-play converges to stable task-level equilibria*

The first question is whether mission-level self-play produces a reproducible endpoint rather than a transient training phase. Figure 2 shows the full three-stage trajectory for S7 seed 20260801. After the entropy schedule was frozen, the five 200-iteration windows from 2000 to 3000 had h2h means 0.3465, 0.3428, 0.3484, 0.3507, and 0.3463. The corresponding jam-vs-sweep means were 0.5121, 0.4956, 0.5118, 0.5107, and 0.5215. This continuation therefore rules out both an early stopping artifact and a persistent drift. The last 40 validation points at 3000 give h2h $0.3485 \pm 0.0151$ and jam-vs-sweep $0.5161 \pm 0.0174$.

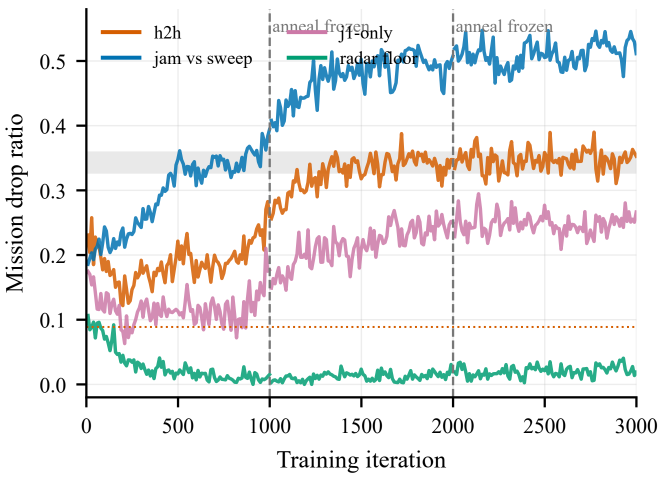


Fig. 2. Three-stage S7 trajectory for seed 20260801. Dashed lines mark the 1000- and 2000-iteration continuation boundaries; the final region is stationary across the four evaluation views.

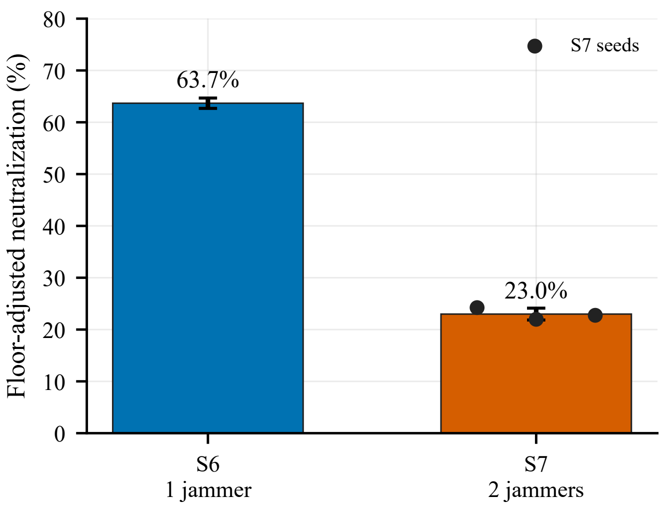


Fig. 3. Floor-adjusted neutralization for the S6 and S7 games. The dots show the three converged S7 training seeds; error bars are standard deviations across training seeds (S6 uses the two valid 12-dB seeds).

The same endpoint appears across three converged S7 seeds. Full-protocol evaluation gives h2h drops of 0.3390, 0.3212, and 0.3496, with an across-seed mean of $0.3366 \pm 0.0143$. The idle-radar drop is $0.0194 \pm 0.0008$ in S7, compared with $0.0275 \pm 0.0110$ across the two valid S6 seeds; both values remain low. The convergence result is descriptive: it establishes a stable attractor for this benchmark and training protocol, not a proof of global Nash equilibrium.

### *B. Doubling the attacker reduces defense containment by two-thirds*

The main comparison is summarized in Table IV and Figure 3. In S6, the valid two-seed 12-dB h2h drop is $0.0888 \pm 0.0053$, jam-vs-sweep is $0.2751 \pm 0.0110$, and floor-adjusted neutralization is $63.7\% \pm 1.0\%$. In converged S7, these values become $0.3366 \pm 0.0143$, $0.5294 \pm 0.0215$, and $23.0\% \pm 1.1\%$, respectively.

TABLE IV
CONVERGED S6–S7 COMPARISON. DROP IS THE MISSION-LOSS RATIO; $\eta$ IS FLOOR-ADJUSTED DEFENSE NEUTRALIZATION.

| Metric | S6: 1v2 | S7: 2v2 |
|---|---|---|
| h2h drop | $0.0888 \pm 0.0053$ | $0.3366 \pm 0.0143$ |
| jam-vs-sweep drop | $0.2751 \pm 0.0110$ | $0.5294 \pm 0.0215$ |
| rad-vs-idle drop | $0.0275 \pm 0.0110$ | $0.0194 \pm 0.0008$ |
| $\eta$ | $63.7\% \pm 1.0\%$ | $23.0\% \pm 1.1\%$ |

TABLE V
EVALUATION-ONLY SCRIPTED BASELINES AGAINST THE CONVERGED S7 TEAMS (MISSION DROP; MEAN $\pm$ SD ACROSS THREE TRAINING SEEDS, THREE ACTION SEEDS EACH). EDF IS A DEADLINE-INFORMED ORACLE (PRIVILEGED DEADLINE STATE).

| Policy | Mission drop |
|---|---|
| *Radar side, vs. trained jammer pairs* | |
| Learned radar team (h2h) | $0.3366 \pm 0.0143$ |
| Scripted sweep | $0.5294 \pm 0.0215$ |
| Random legal radar | $0.5085 \pm 0.0051$ |
| Greedy mission-stare radar | $0.0889 \pm 0.0000$ |
| EDF oracle radar | $0.2961 \pm 0.0039$ |
| *Jammer side, vs. trained radar teams* | |
| Learned jammer pair (h2h) | $0.3366 \pm 0.0143$ |
| Stare jammer pair | $0.2374 \pm 0.0036$ |
| Random legal jammer pair | $0.0922 \pm 0.0047$ |
| Idle jammers (reference) | $0.0194 \pm 0.0008$ |

The neutralization change is not a consequence of a corresponding radar collapse. Learned radars facing idle jammers retain a low drop of $0.0194 \pm 0.0008$ in S7, compared with $0.0275 \pm 0.0110$ across the two valid S6 seeds. Instead, two jammers increase the marginal disruption that survives radar adaptation. The jammer team's raw disruption against scripted sweeps rises by 92% relative to S6, while the fraction of that disruption removed by learned radars falls by approximately two-thirds. Because the comparison controls total jammer activation but also changes radar-site geometry, this is a benchmark-level scaling result rather than a fully isolated attacker-count intervention.

### *C. Scripted baselines anchor both teams and expose co-adaptation*

Table V evaluates the converged teams against four scripted policies under the identical 64-scenario, three-action-seed protocol. Both learned teams dominate their non-learning counterparts: a random legal radar loses $0.5085 \pm 0.0051$ of missions against the trained jammer pairs versus $0.3366 \pm 0.0143$ for the trained radar team, and random or stare jammers impose only $0.0922 \pm 0.0047$ and $0.2374 \pm 0.0036$ against the trained radars versus $0.3366 \pm 0.0143$ for the trained jammer pairs. Learning is real on both sides of the link.

The greedy mission-stare radar is the informative exception. Pointing both heads at the heaviest pending service–azimuth cell yields a drop of exactly 0.0889 against every trained jammer pair and every action seed—a factor of 3.8 below the learned h2h drop and below even the natural scheduling floor of 0.1187 (sweep radars facing idle jammers). Under this link budget, the pointing gain at the exact mission bearing creates a detection margin that none of the three trained jammer teams overcomes: their learned advantage applies to radars that spread attention (sweeps, $0.5294$; the learned equilibrium radar, $0.3366$), not to a well-pointed stare. The ordering among heuristics is itself diagnostic: the hottest-queue stare ($0.0889$) dominates a deadline-first oracle ($0.2961 \pm 0.0039$) despite the latter's privileged deadline access, because per-step detection is probabilistic and grinding the densest pending sector converts more expected detections than chasing the single earliest deadline.

This reframes the h2h value and $\eta$: they are equilibrium quantities, not radar performance ceilings. The self-play radar hedges against its co-trained opponent at the cost of mission throughput, and the learned jammer team is exploitable by an out-of-distribution non-learning policy—the radar-side counterpart of the singleton result in Section V-D. A jammer team co-trained against greedy radars would plausibly restore the penalty; testing that counter-adaptation requires new training runs and is left as future work.

### D. Attacker count is primary; cross-fire geometry is secondary

The co-located-jammer control tests whether the S7 result requires two spatially separated jammers. It places both jammer sites at $+60°$ while retaining the two radar sites at $\pm 20°$, the same physics, budget, two-stage training, and final evaluation. The corrected control yields h2h $0.2927 \pm 0.0119$, jam-vs-sweep $0.4947 \pm 0.0026$, and neutralization $28.4\%$.

Thus, the matched S7 geometry control shows that two jammers account for most of the containment loss within the separated-radar setting: containment is $28.4\%$ for co-located jammers and $24.2\%$ for cross-fire. The S6-to-S7 comparison additionally changes radar-site geometry, so the full count effect is a benchmark-level attribution rather than a perfectly isolated causal contrast. The cross-fire comparison itself is controlled for radar placement, budget, training protocol, and evaluation. The pre-specified contestability profiles and the matched control are shown in Figure 1; the converged mechanism decomposition is shown in Figure 4.

### E. Pair-trained defenses trade singleton robustness for pairwise defense

The fourth view evaluates one learned jammer against radars trained in the 2v2 game. Across the three converged S7 seeds, the singleton drop is 0.2632, 0.2387, and 0.1238, with mean $0.2086 \pm 0.0745$. This variation is larger than the h2h variation, but all three seeds expose the same opponent-class boundary: a defense optimized against a pair is not uniformly robust to a singleton.

The seed-01 continuation makes the trade-off visible over time. As the pair-trained equilibrium develops, j1-only rises from approximately 0.12 near the 1000-iteration checkpoint to approximately 0.24 at 2000 and remains near 0.25 at 3000, while h2h settles near 0.34. Radar competence against idle jammers remains low and stable. The result is therefore an adaptation trade-off, not evidence that the radar actor or critic stopped learning.

TABLE VI
R5-LITE OPPONENT-CLASS MIXTURE. ABSOLUTE DROPS ARE DESCRIPTIVE BECAUSE SINGLETON ITERATIONS SKIP THE JAMMER UPDATE; RATIO METRICS ARE THE PRIMARY COMPARISON.

| $q$ | h2h | jvs | j1-only | j1/jvs | $\eta$ |
|---|---|---|---|---|---|
| 0 | .3282 | .5043 | .2532 | .502 | 20.2% |
| .25 | .2470 | .4015 | .1767 | .440 | 23.5% |
| .50 | .1962 | .3580 | .0992 | .277 | 26.1% |
| .75 | .1939 | .3476 | .0911 | .262 | 24.6% |

The learned behavior is interpretable and is summarized in Figure 5.

### F. Opponent-class mixing reduces singleton leverage

The full dose-response is shown in Figure 6 and Table VI.

Singleton relative leverage $d_{\mathrm{j1}}/d_{\mathrm{jvs}}$ falls monotonically from 0.502 to 0.262 and saturates between 50% and 75% exposure. The containment ratio $\eta$ increases from 20.2% to 26.1% at 50% exposure before leveling at 24.6%. The h2h-to-jvs ratio is $0.651$ at $q = 0$, falls to $0.548$ at $q = 0.50$, and is $0.558$ at $q = 0.75$. It therefore improves through the $50\%$ condition and changes little thereafter; the ratio-level data do not show a pairwise-versus-singleton trade-off in this intervention.

R5-lite has an important design boundary. On singleton iterations, jammer 1 is forced idle and its update is skipped; consequently the jammer team receives fewer gradient updates as $q$ increases, and absolute jvs drops from 0.5043 to 0.3476. The dose-response does not identify a causal improvement in absolute radar performance. It does support a narrower intervention claim: under this minimal mixture and fixed training budget, exposing the radar team to a singleton opponent reduces the singleton's relative leverage, with diminishing returns after 50% exposure. A gradient-aligned multi-seed experiment is needed before calling the result universal.

## VI. Discussion

The experiments reveal a distinction that is hidden by link-level reporting. Increasing the attacker team from one to two does not make self-play unstable; it changes how much of the jammer's marginal disruption the learned radar team can remove. Under the tested task workload, the same radar competence floor coexists with a much larger adversarial loss. The relevant design variable is therefore not only the quality of a single radar policy, but also the number and spatial arrangement of simultaneous interference sources.

The scripted baselines add a second distinction: an equilibrium policy is not a mission-optimal policy. The learned jammer teams cannot punish a greedy mission-stare radar (drop 0.0889, invariant across all three trained teams), while they punish sweeps ($0.5294$) and the learned radar ($0.3366$). Both sides of the self-play game are specialized to the training opponent distribution: the radar gives up mission throughput to hedge against its co-trained jammer, and the jammer's disruption is effective precisely against attention-spreading radars. Head-to-head drop and containment $\eta$ must therefore be read

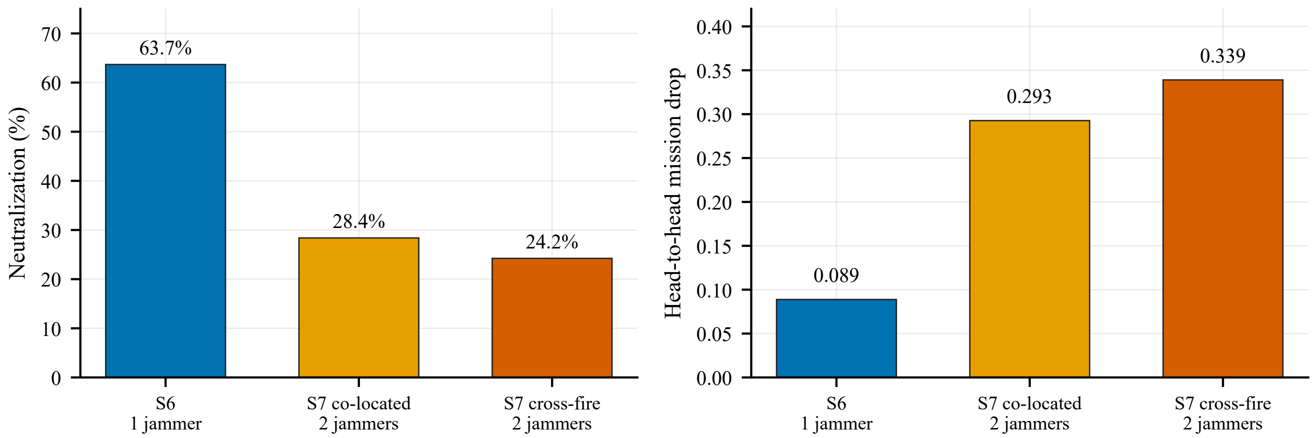


Fig. 4. Mechanism decomposition under matched task, budget, and training protocols. Two co-located jammers reproduce most of the containment loss, while cross-fire geometry adds a secondary penalty.

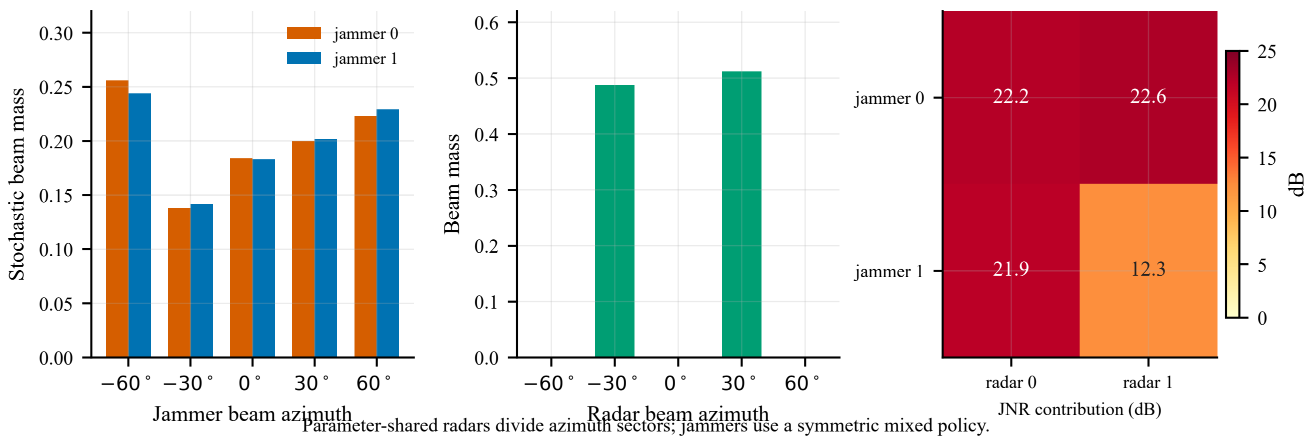


Fig. 5. Equilibrium behavior at the converged S7 checkpoint. Radar heads divide the mission plane between two azimuth sectors, while the jammer team uses a symmetric mixed beam policy. The heat map reports active-only pairwise JNR contributions in dB.

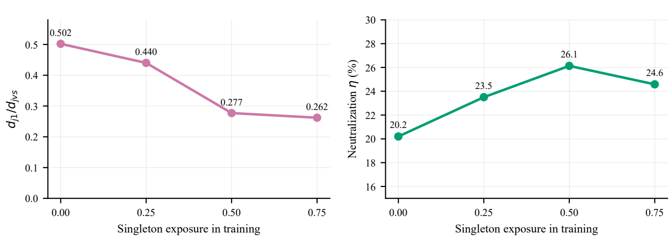


Fig. 6. R5-lite opponent-class mixing. Singleton relative leverage $d_{j1}/d_{jvs}$ falls with singleton exposure, while floor-adjusted neutralization peaks near 50%. Absolute drops are confounded by skipped jammer updates.

as properties of the co-adapted pair, not as absolute radar or jammer ratings. This is the same lesson as the singleton-vulnerability result, seen from the other side of the game: static evaluation against out-of-distribution opponents—even non-learning ones—is the cheapest robustness probe available, and it should accompany any self-play claim.

The mechanism control refines this statement. Two jammers at one bearing already reduce containment from 63.7% to 28.4%. Cross-fire reduces it further to 24.2%. The benchmark therefore assigns primary responsibility to attacker count and a secondary penalty to geometry. This decomposition is more useful than attributing the entire effect to a visually appealing cross-fire explanation: a defense that survives one jammer should be stress-tested against the energy and timing consequences of two sources even when their directions are similar.

R5-lite suggests a practical training response. Mixing opponent classes reduces singleton relative leverage, and the gain saturates near equal exposure to pair and singleton opponents. The result motivates opponent-distribution design as a complement to longer self-play. It does not imply that a full AlphaStar-style league is required for the S7 equilibrium itself [24]. In this benchmark, single-policy self-play converges after approximately 1700 iterations. A population or league becomes relevant when the design objective includes robustness to threat compositions that were not present in the training game.

The pre-training contestability gate provides a cheap way to decide whether a proposed physical regime can support such a study. It tests whether the best radar response spans a useful probability interval before expensive optimization begins. In S7, the gate predicted that cross-fire would reduce selected sectors while keeping the game nontrivial; the trained behavior and co-located control followed that direction. This gate is a design diagnostic, not a substitute for training or a proof of equilibrium.

The physics contract is deliberately minimal, and the direction of each simplification can be stated. Broadband noise jamming with rectangular spectra is the canonical denial waveform; smarter coherently-informed jamming would only strengthen the attacker, so the attacker-side conclusion is conservative, as is the incoherent cross-jammer power summation argued in Section III. Idealized phase-aligned cell aggregation within one jammer slightly favors the jammer over a calibrated real array. The exclusions cut in the opposite direction: real radars retain Doppler/MTI filtering and waveform agility that our radar agents do not exercise, so the defense as modeled is weaker than a fielded system, and the h2h values measure scheduling policy under the stated budget rather than absolute electronic-protection capability. The coarse $5 \times 5$ beam grid quantizes pointing symmetrically for both sides. Within this contract the conclusions are exact; the running SNR-regime sweep tests their budget sensitivity, and geometry and waveform-family breadth remain open.

## VII. Limitations

First, the physical model uses a single five-sector azimuth grid, two services, fixed site bearings, and a simplified broadband-jammer link budget. The count–geometry decomposition therefore applies to the tested geometry family; it is not a universal scaling law for deployed arrays. The greedy-stare result is likewise a property of this link budget: a stronger jammer or a smaller pointing margin could close the stare gap, and the experiment measures the trained teams only against static baselines, not against jammers co-trained to exploit a stare. Second, the study is simulation-only. Mission drop is a model-level operational endpoint and does not establish fielded detection, tracking, or electronic-support performance. Third, the main S7 equilibrium has three training seeds, but the co-located mechanism control uses one training seed and R5-lite uses one seed per mixture condition. Fourth, R5-lite skips the jammer update on singleton iterations, so its absolute drop values confound opponent exposure with jammer gradient budget. The ratio-level result is the intended readout, while gradient-aligned multi-seed mixing remains necessary for a stronger causal intervention claim.

## VIII. Conclusion

We presented a task-level benchmark for adversarial co-adaptation between multifunction radars and cooperative jammers. The benchmark links array-factor physics and finite energy to bearing-specific missions with deadlines, then evaluates both sides with controls that separate natural scheduling loss from adversarial loss.

Across three converged seeds, adding a second jammer reduced learned-radar containment from $63.7\% \pm 1.0\%$ in the valid two-seed S6 baseline to $23.0\% \pm 1.1\%$ at the same S7 team budget, while idle-jammer radar loss remained low. A co-located control showed that two jammers under the separated S7 radar geometry already produce most of the reduction, with cross-fire adding a smaller penalty. Longer continuation training confirmed a stationary high-loss equilibrium rather than a non-transitive training cycle. Finally, a lightweight opponent-class mixture reduced singleton relative leverage with diminishing returns near 50% exposure.

Three engineering insights transfer beyond this specific benchmark. First, attacker count is the primary stress variable: a defense validated against one jammer should be re-tested against two co-located sources before geometry is even considered, because most of the containment loss in our matched control came from count alone. Second, an equilibrium policy is not a mission-optimal policy: learned schedulers hedge against their training opponent at the cost of task throughput, so any self-play claim should be accompanied by cheap static evaluations against out-of-distribution opponents—including non-learning heuristics, which exposed the jammer side here as clearly as the singleton exposed the radar side. Third, pointing discipline is the strongest single lever in the modeled budget: mission-stare margins survived every trained jammer team, which suggests that guaranteeing a per-mission pointing margin deserves at least as much design attention as learning the scheduler built on top of it. The next rigorous extensions are a gradient-aligned, multi-seed mixture study, a broader geometry and SNR family, and counter-adaptation training against the stare heuristic, followed by hardware-in-the-loop validation.

## Reproducibility

The complete evaluation pipeline is deterministic and self-contained: fixed train and validation scenario manifests, fixed action seeds, atomic checkpoints, 18 automated environment and physics gates, and a results-integrity script that re-derives every number quoted in this paper from the raw evaluation JSON files and fails on any mismatch. Upon acceptance, the authors will deposit as supplementary material: (i) the environment, trainer, and evaluation source code; (ii) the training and validation manifests; (iii) all final-evaluation and baseline JSON results for every reported seed; (iv) the figure-generation scripts and the single-source results table; and (v) the three converged checkpoints, so that every table and figure in the paper can be regenerated with one command per figure. A public repository with a citable DOI (IEEE DataPort or equivalent) will accompany the final version; the frozen internal revision identifier is retained in the project log until then.

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